GOOGLE CLOUD AI BUILDER CUP 2026 · BFSI TRACK

Property Inspection AI

From a 20-second phone video to an underwriting-ready property risk file

Defects, condition grade, repair cost, valuation impact, fraud signals and a 3D proof of what was inspected, built for mortgage lenders, insurers and property inspectors.

Gemini 3.8 Flash · Gemma 4 · Gemini API · Cloud Run + L4 · Cloud Storage · Cloud Build



Concept illustration of the product

Team Xcore · Track: Intelligent Risk, Fraud & Financial Experiences · Prototype v1.0.0 · October 2026

THE TEAM

Team Xcore

Three builders covering Gen AI, vision and cloud architecture, from research to a deployed product.



Thanveer Ahamed

Gen AI Lead

Gemini and Gemma reasoning, prompts, verification, analysis and the underwriting assistant



Janitha Prathapa

Vision AI Lead

Defect detection, Depth Anything 3, SLAM, 3D model, floor plan and coverage map



Dinal Fernando

Solution Architect

Google Cloud architecture: Cloud Run service + GPU Job, Cloud Storage and deployment

THE PROBLEM

Property risk is judged from photos nobody can verify

$308.6B

estimated yearly US insurance fraud, all lines

Coalition Against Insurance Fraud, 2022

~$45B

of that in property & casualty insurance

same study

$4.5–12.5B

global home-inspection services market, 2025

industry market reports

Slow, costly site visits

Lenders and insurers wait days for an inspector; remote cases go unchecked.

Unverifiable self-reported photos

Photos get reused, edited, shot off a screen or taken at another property.

No proof of what was inspected

Nobody can tell which walls or ceilings a remote inspection actually covered.

Subjective, inconsistent grading

Two inspectors grade the same room differently, and the reasons go unrecorded.

Collateral overvaluation

Regulators such as the RBI have flagged inflated valuations in housing-finance fraud.

WHO IT SERVES

One inspection record, five BFSI decision-makers

Mortgage lenders, HFCs & NBFCs

Collateral condition grade, valuation impact %, repair retention and a REFER flag before disbursal and over the loan's life.

Insurers: underwriting

Insurance perils, risk level and a premium-loading suggestion at quote and renewal, without a site visit for low-risk homes.

Insurers: claims

Claim-ready evidence with an indicative repair-cost range, plus recycled-video and screen or AI-footage detection.

Inspectors & valuers

The AI drafts the report and the coverage map shows what to film again, so one inspector can supervise many remote inspections.

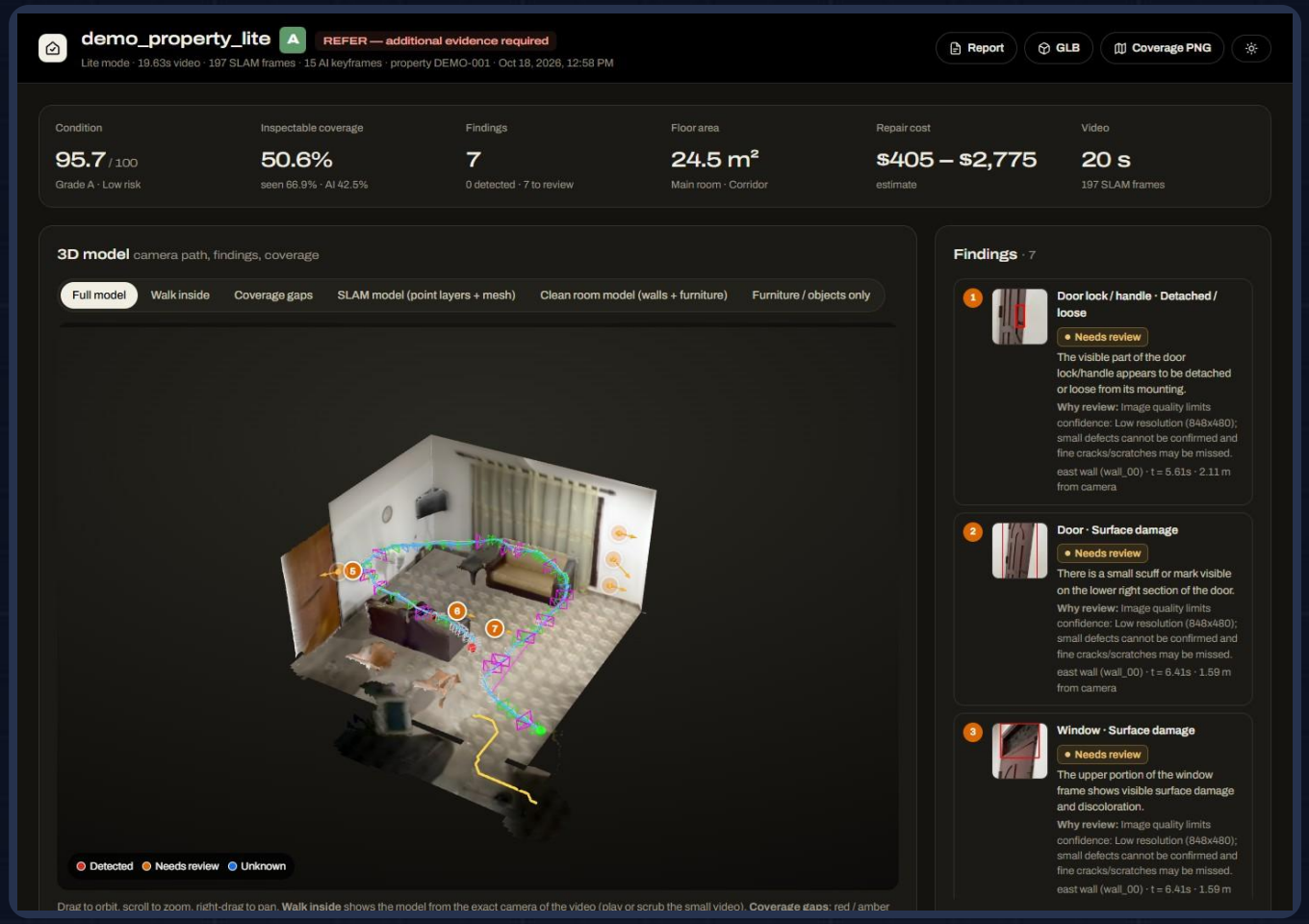
Real estate, owners & buyers

A transparent condition report, 3D model and floor plan, with fraud-resistant evidence for listings and sales.

Same evidence, one record: every inspection is a JSON record behind a REST API that each team's own system can use

THE SOLUTION

One upload gives you a complete, explainable risk file



Live app: walkthrough report with the 3D model, findings, condition grade and underwriting decision

- Defects localised and verified
- Condition grade A–E (0–100 score)
- Risk level and insurance perils
- Indicative repair-cost range
- Valuation and collateral impact
- Premium-loading suggestion
- Underwriting decision with reasons
- Evidence-integrity (fraud) flags
- 3D model, floor plan, coverage map
- Gemini analysis and underwriter Q&A

HOW IT WORKS

Capture, inspect, decide, act: minutes instead of days

1 Capture

Walk through the property with a phone and upload the video. No special hardware.

2 AI inspection

On each AI keyframe, Gemini reads the room, LocateAnything finds candidate defects and Gemma 4 verifies each one.

3 Risk decision

A transparent rule engine sets the grade, perils, repair cost, valuation, premium and decision; Gemini explains it.

4 Act

HTML/PDF-ready report, Gemini Q&A and a JSON record over the REST API for underwriting and claims systems.

10.6 min

end to end for a 20 s video, laptop GPU

≈ $0.85

cloud cost per walkthrough, L4 Job + Gemini

≈ $0.35

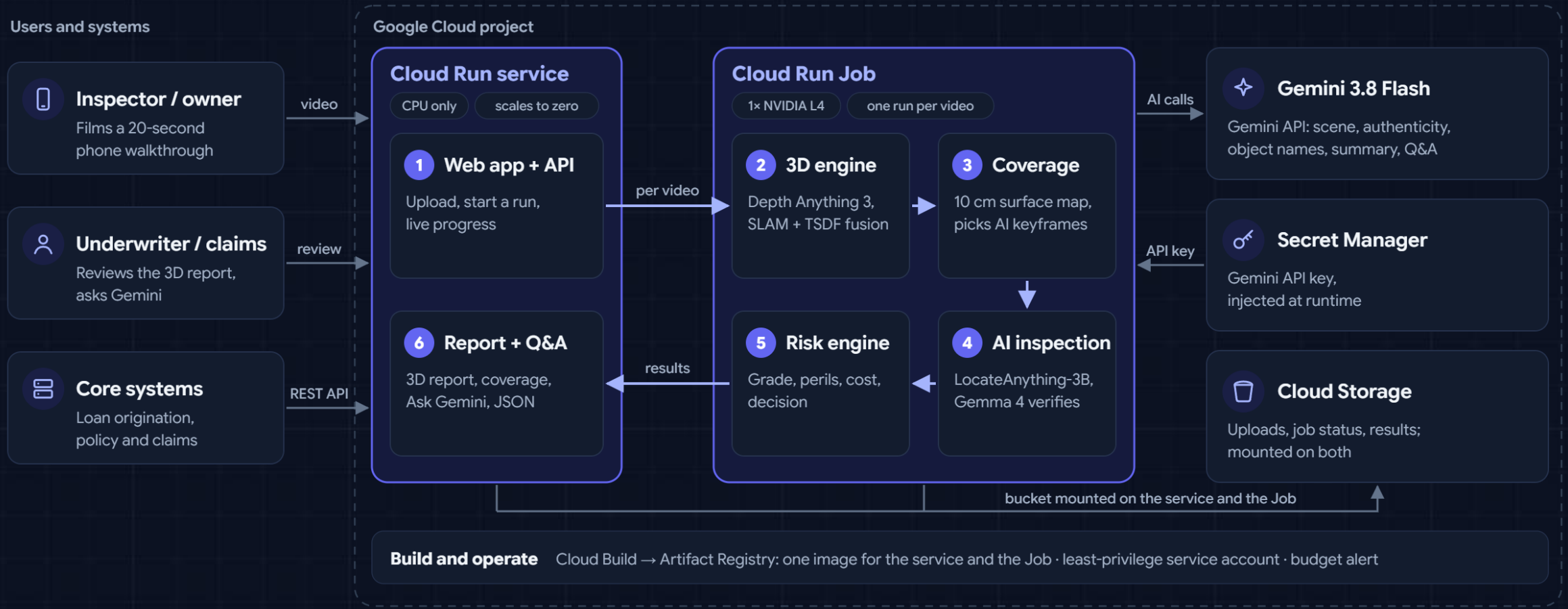
Gemini cost per walkthrough

$0

GPU cost when idle: the L4 Job runs only per video

SOLUTION ARCHITECTURE

Cloud Run web service + L4 GPU Job, powered by Gemini



Deployed layout (v1.0.0): web UI + API on a CPU-only Cloud Run service; 3D engine, coverage and AI inspection on a Cloud Run Job with one NVIDIA L4 per video; a Cloud Storage bucket mounted on both for uploads, status and results; Gemini 3.8 Flash via the Gemini API (key in Secret Manager). BigQuery and Pub/Sub are not used.

AI MODELS

Hybrid AI: the right Google and open model for each job

Job	Model	Runs on	Why this model
Room, visible components, keyframe authenticity	**Gemini 3.8 Flash**	Gemini API	Strong visual reasoning and reliable JSON output
Find candidate defects (boxes)	**LocateAnything-3B**	Cloud Run GPU Job	High-recall open-vocabulary localisation
Verify each defect on a full-resolution crop	**Gemma 4 E4B**	Cloud Run GPU Job	Open Google model, cheap per box, cuts false positives
DETECTED / NEEDS_REVIEW / NOT_DETECTED	**Rule engine + Gemini**	CPU + Gemini API	Rules decide, so the result is auditable; Gemini explains each finding
Grade, perils, repair cost, valuation, premium : risk_rules.yaml	**Rule engine + Gemini**	CPU + Gemini API	Rules are tunable per lender or insurer; Gemini writes the risk summary
Depth and 3D model from a phone video	**Depth Anything 3 + OpenCV 5 SLAM**	Cloud Run GPU Job	Room-scale 3D from a single phone camera
3D object names, summary, analysis, Q&A	**Gemini 3.8 Flash**	Gemini API	Grounded on the inspection record and cites issue IDs

KEYFRAME PIPELINE

Find broadly, verify with a second model, then decide

1. Quality check + integrity (dHash fingerprint)
2. Gemini: room, components, authenticity
3. Prompt router: 18 defect types, 12 components
4. LocateAnything-3B boxes → NMS / merge
5. Full-res crops → Gemma 4 verify + CV evidence
6. Rule engine → status, then risk engine









Real outputs: wall crack, corridor peeling paint, bathroom water damage, walkthrough keyframe (door, window)

DETECTION SCOPE

What we detect: 18 visible defect types on 12 components

Wall

Crack, stain, water damage, peeling paint, hole, mold, surface damage, discoloration, scratch

Ceiling

Crack, stain, water damage, peeling paint, hole, mold

Floor tile

Crack, chip, missing tile, stain

Wall tile

Crack, chip, mold, stain

Grout

Crack, missing grout, discoloration, mold

Wood / vinyl floor

Scratch, stain, water damage, damaged boards

Countertop

Scratch, crack, chip, stain, burn mark, discoloration

Door

Surface damage, hole, crack, scratch, chip, peeling paint

Door lock / handle

Missing, broken, rust, loose

Window

Broken or cracked glass, frame damage, wet sill, mold

Molding / trim

Chip, crack, detached, water damage, missing

Kitchen exhaust

Grease buildup, rust, broken grille, missing filter, dents



Professional review

Cracks and deformation get a structural note; water damage and mold get a moisture-source check.



Manual checks, never AI

Doors and windows opening, locks and keys, exhaust fan and motor.



Look-alikes rejected

The verifier is shown grout lines, wood grain, shadows, reflections and outlets.



Not inspected

Furniture, cupboards, rugs and carpets.

EXPLAINABLE RISK ENGINE

Every decision shows its reasons



Live app: property risk & valuation panel for the case-study walkthrough

Condition score

100 – Σ (points × severity × state weight)

A	≥85	B	≥70	C	≥50	D	≥30	E	<30

Underwriting decisions

- PROCEED
- PROCEED WITH CONDITIONS: note defects
- PROCEED WITH CONDITIONS: repairs / retention
- REFER: professional survey required
- REFER: evidence integrity
- REFER: additional evidence required

FRAUD & EVIDENCE INTEGRITY

Fraud checks run on every walkthrough video

Recycled video

Every AI keyframe is fingerprinted (dHash) and compared with every property. A match to another property is flagged HIGH.

Screen, render or AI footage

Gemini checks each keyframe for a screen recording, a rendering or AI imagery.

Low-quality frames

Blurred frames are dropped; low-resolution keyframes stay Needs review.

Traceable evidence

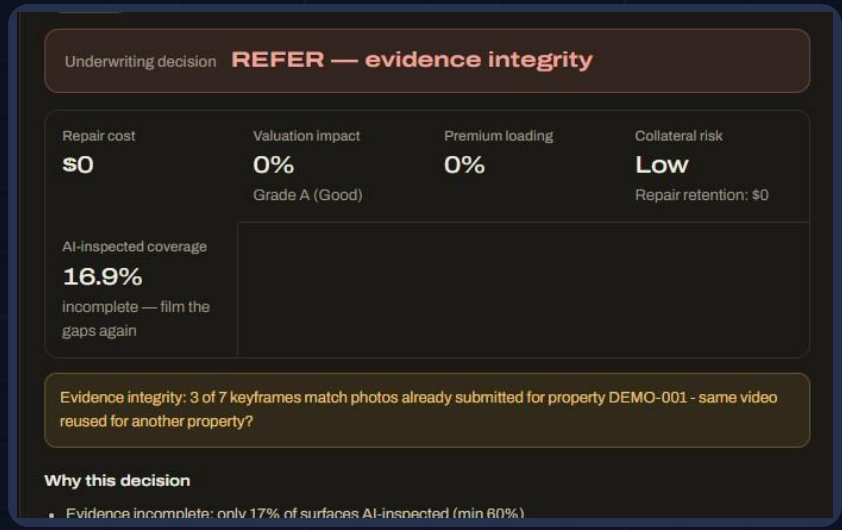
Every finding keeps its keyframe, close-up, wall and video time.

Unfilmed areas

The coverage map names each wall and ceiling the video missed, to film again.

Incomplete video coverage

Under 60% AI-inspected surface → REFER: additional evidence required.

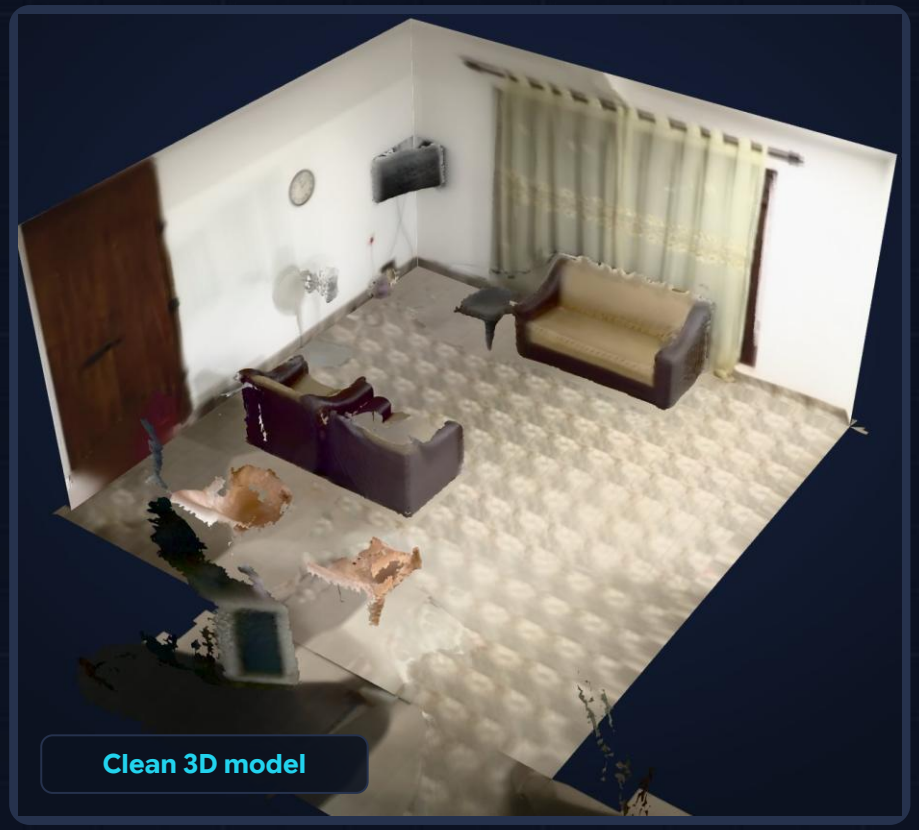


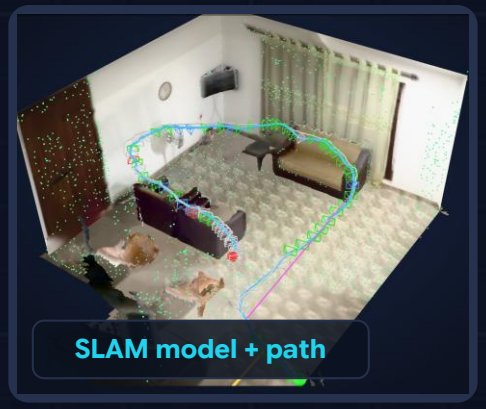
Any HIGH flag sends the case to REFER: evidence integrity

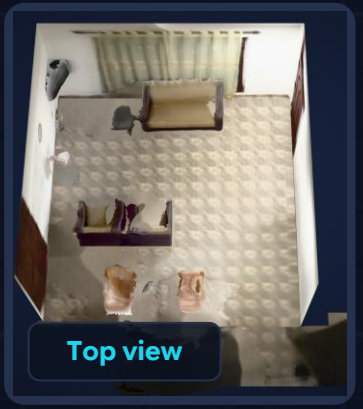
A human reviews the case before it can go ahead: it stops recycled-video and wrong-property claims and loan files.

VIDEO → 3D DIGITAL TWIN

A phone video becomes a 3D model and floor plan

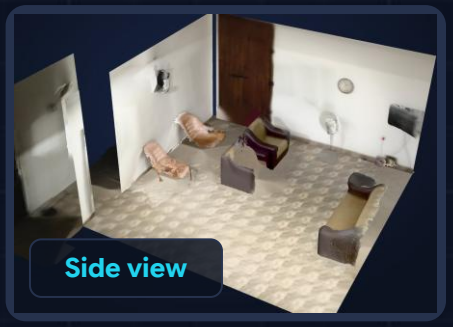


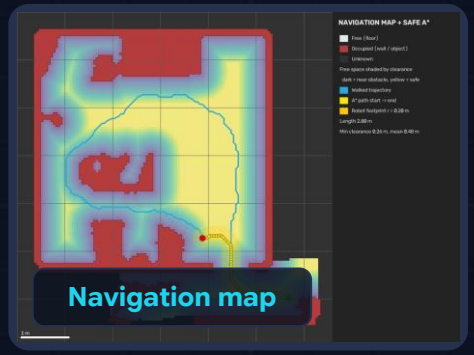










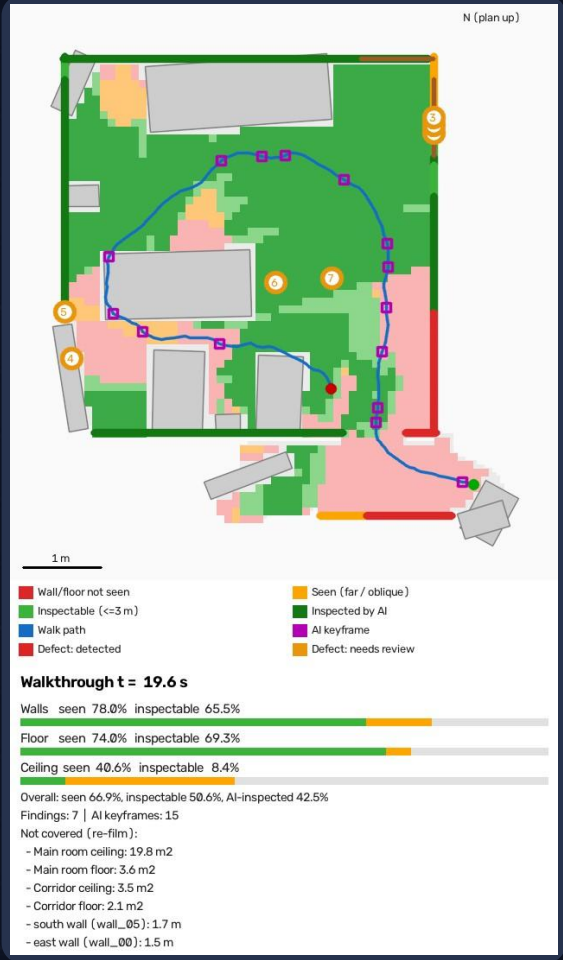


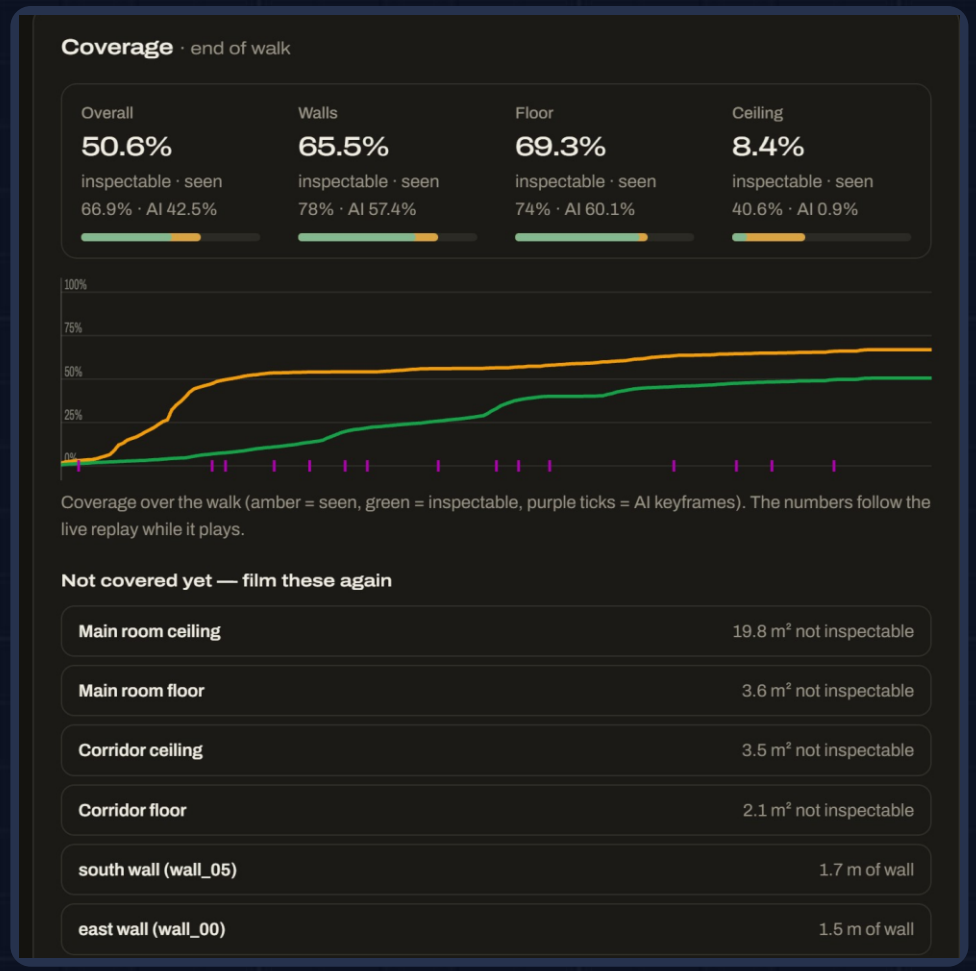
Depth Anything 3 depth per frame → **OpenCV 5 SLAM** poses → **Open3D TSDF** surface → walls, floor, ceiling, doors → **objects named by Gemini**

Outputs: clean and SLAM GLB models, floor plan (PNG + SVG) with room areas, occupancy and navigation maps, in approximate metres

COVERAGE MAP: OUR KEY DIFFERENTIATOR

Proof of what was inspected, and what to film again





Every 10 cm of surface

is checked against every frame's depth: seen, inspectable (close and frontal) or AI-inspected.

Smart AI keyframes

A greedy set cover picks the fewest frames that cover the inspectable surface.

Findings pinned in 3D

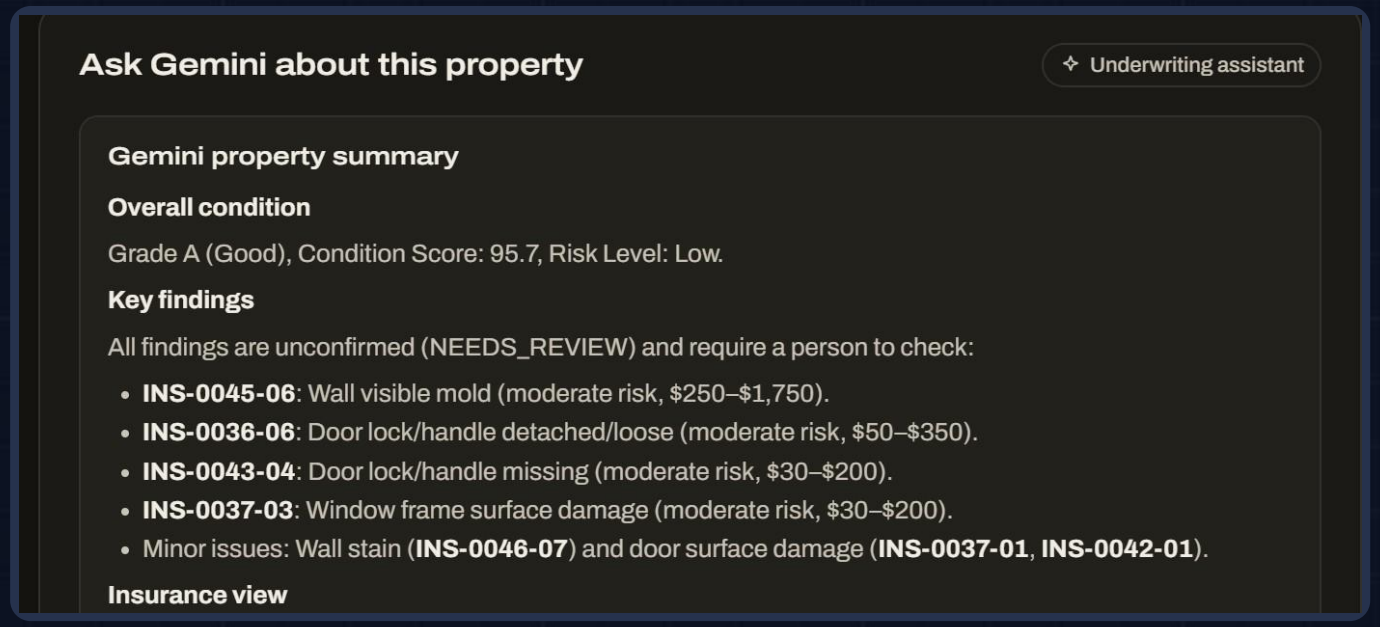
Each defect gets a wall and a time in the video, and is merged across frames.

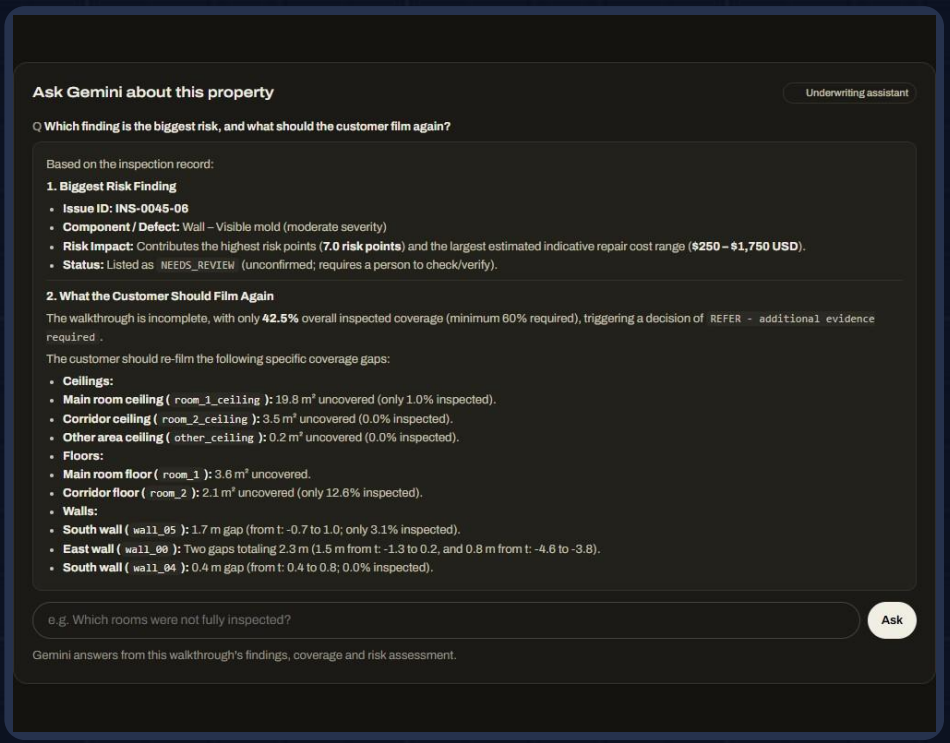
Re-film list

Below 60% AI-inspected → REFER, with the exact walls and ceilings to film again.

GEMINI 3.8 FLASH

Gemini works as the underwriter's co-pilot





Property analysis

Summary, causes, action plan, insurance and lending views, and site-visit questions

Grounded Q&A

Answers only from this inspection's findings, coverage and risk, and cites issue IDs

Scene and authenticity

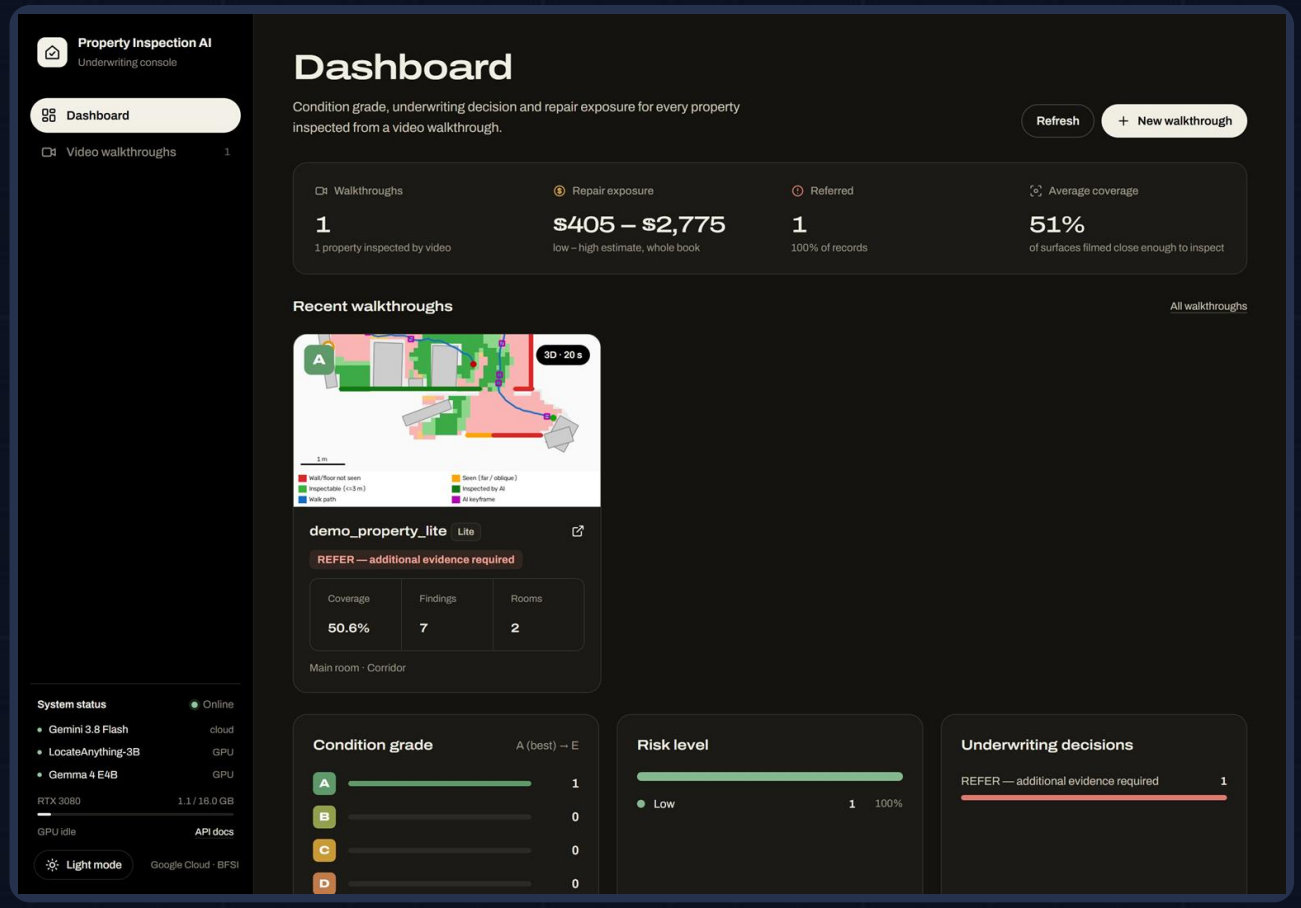
Room type, visible components, and screen, render or AI-image detection

3D object naming

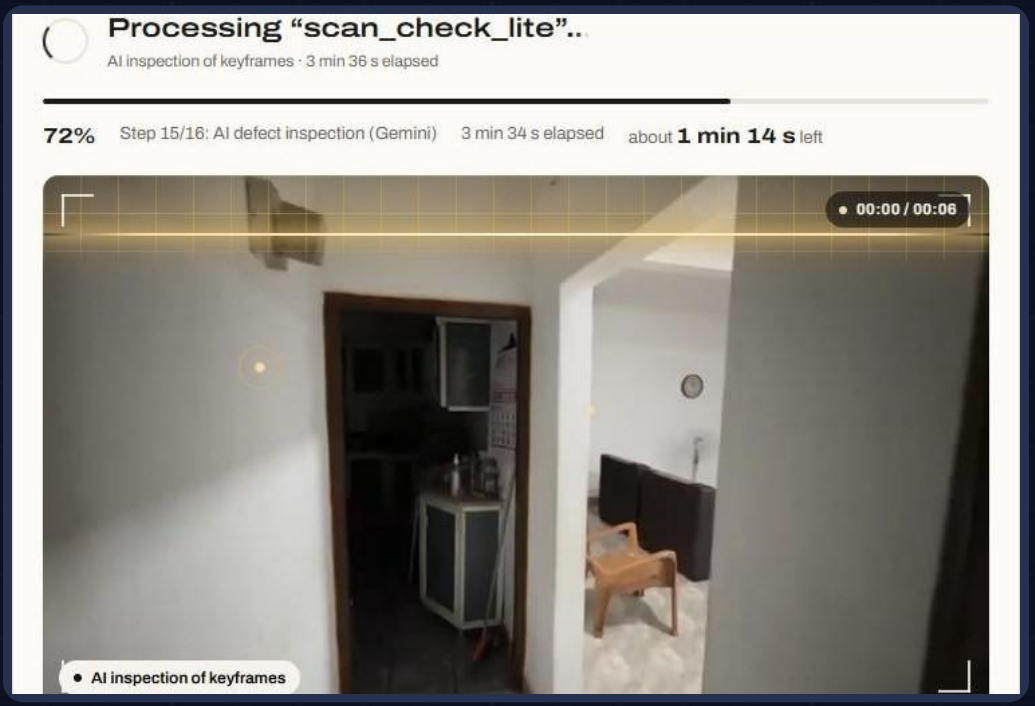
Names the furniture and fixtures segmented from the 3D model

PRODUCT EXPERIENCE

A product built for underwriters



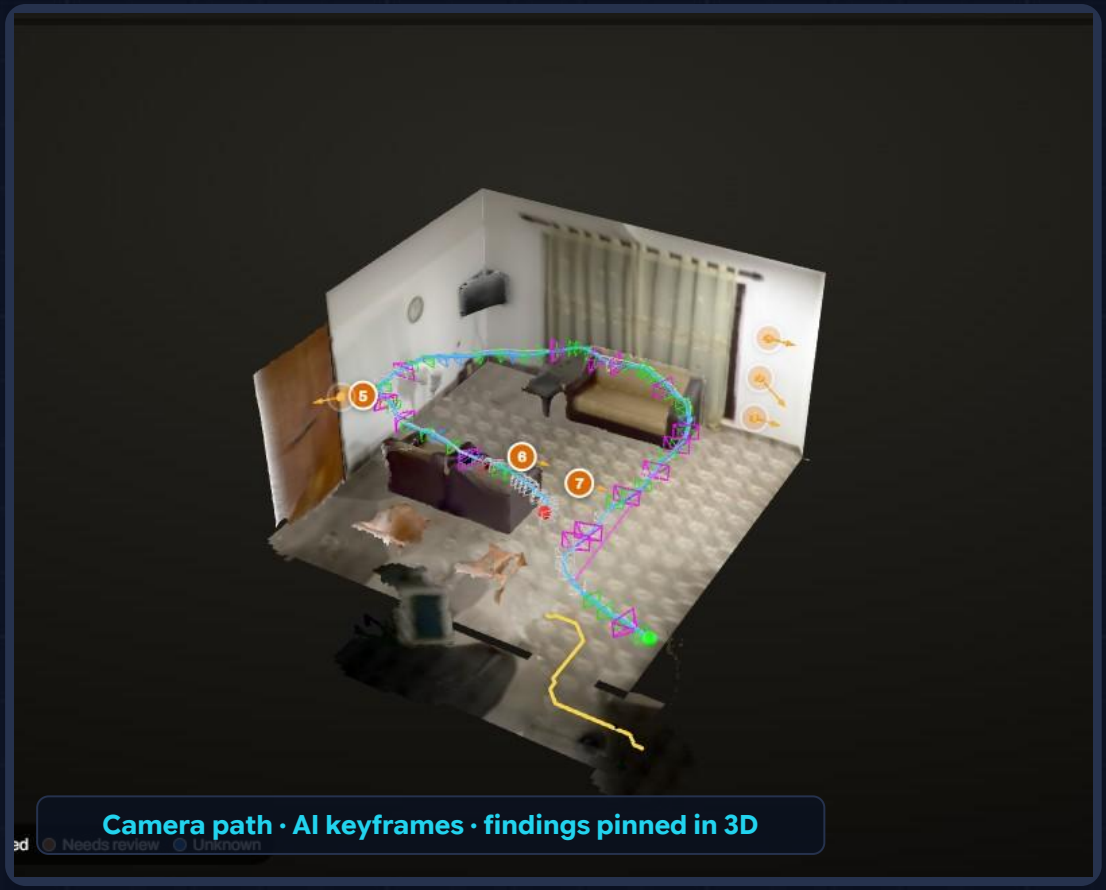
Dashboard: walkthroughs, repair exposure, referrals and coverage across the whole book



- Video upload, live 3D-scan view, Lite or Advanced
- Interactive 3D viewer: walk inside, coverage gaps
- Reports in HTML, Markdown and JSON; REST API at /docs
- Light and dark themes; live video-queue status

CASE STUDY

A real 20-second home walkthrough, end to end



19.6 s
phone video

197
SLAM frames

15
AI keyframes

24.5 m²
2 rooms measured

95.7 · A
condition score

$405–2,775
indicative repairs

Decision: REFER, additional evidence required

The home looks fine, but only 42.5% of its surfaces were AI-inspected (minimum 60%), so the app lists the ceilings and walls to film again. It does not over-promise.

7 possible defects pinned in 3D, all Needs review · 10.6 min end to end on a laptop RTX 3080 · 3D build Lite 11.1 vs Advanced 26.9 min (RTX 3050)

EVALUATION DESIGN

A fair test of four pipelines: 10 rooms, 100 regions

A — Gemini only

Baseline: Gemini inspects the full room image.

B — LocateAnything + Gemini

Localise candidates, then Gemini assesses each one.

C — Gemma 4 + Gemini

Gemma 4 checks the image, then Gemini gives the final assessment.

D — All three models

LocateAnything localises, Gemma 4 verifies each crop, Gemini decides. Our pipeline.

Test set

10

rooms: 5 living rooms + 5 bedrooms

2

clean control rooms (one of each)

100

annotated regions per pipeline

40 / 60

defective / non-defective regions

Fair procedure

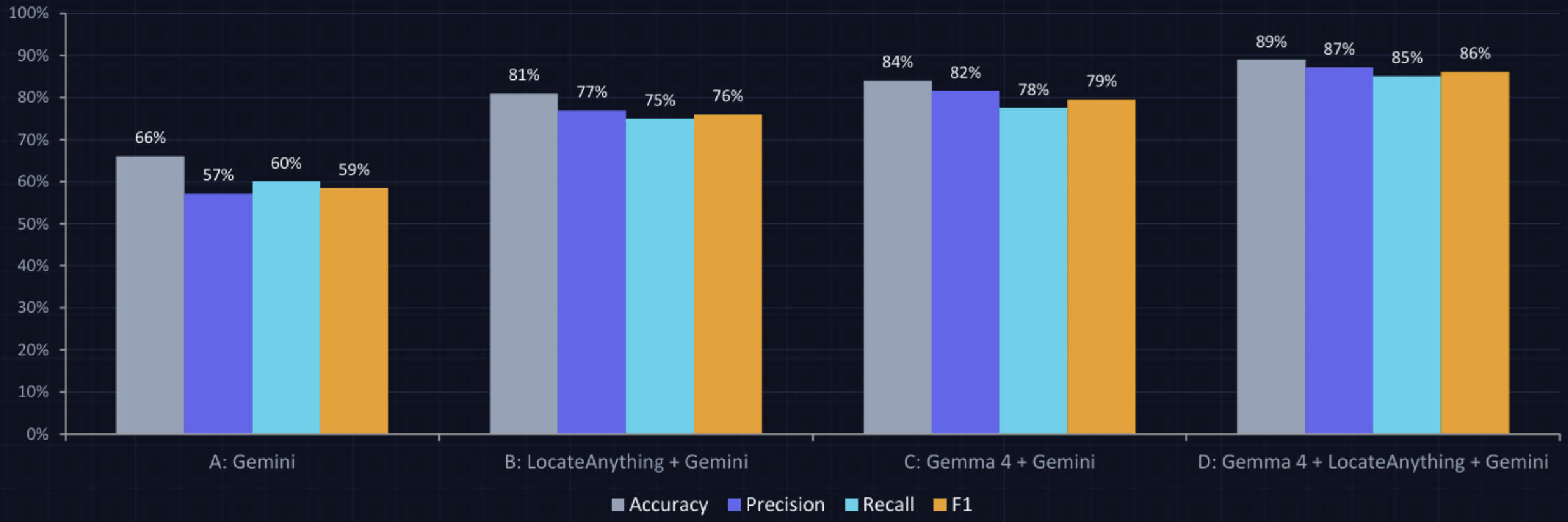
- Same images, taxonomy, output schema and decision rule
- Same Gemini version and settings for every pipeline
- Test set frozen before inference; no labels given to models
- Raw outputs, failures, latency and cost all logged

Acceptance targets for D

- Region accuracy of at least 85%
- Precision, recall and F1 each at least 85%
- At most 1 false alarm on the 2 clean rooms
- Better than Gemini alone (A) on every metric

EVALUATION

Expected gain from adding localisation and verification



clean-room false alarms: A 2 · B 2 · C 2 · D 1 (2 clean rooms, 20 regions)

WHAT WE BUILT

From depth research to a deployable product

01

Depth + SLAM

Depth Anything 3 visual SLAM: trajectory, dense map, occupancy grid, A* path

02

Hybrid SLAM

OpenCV 5 tracking + ICP alignment and Manhattan planes: straight walls, sharp corners

03

Photo-real 3D

Gaussian Splatting experiment (gsplat) on the same cameras

04

Production 3D

One command: video → clean GLB, SLAM GLB, floor plan (+SVG), evaluation

05

Inspection AI

Defect taxonomy, locate + verify models, rule and risk engine, fraud checks

06

Google Cloud v1.0.0

Gemini API, Cloud Run service + L4 Job, Cloud Storage, one-command deploy

18
defect types, 12 components

8
room types

18
REST API endpoints

2
3D modes (Lite / Advanced)

1
command to deploy on Google Cloud

GOOGLE TECHNOLOGIES

Built on Google AI and Google Cloud

IN THE PROTOTYPE

Gemini 3.8 Flash

Via the Gemini API: scene, authenticity, analysis, grounded Q&A, 3D object names

Gemma 4 (E4B)

Open Google model that verifies every defect crop on the GPU

Cloud Run service + L4 Job

One image: CPU web service, L4 GPU Job per video; both scale to zero

Cloud Storage

Bucket mounted on both: uploads, job status, reports, 3D models

Secret Manager

Gemini API key injected at runtime; no keys in the image or repo

Cloud Build

Builds the ~12 GB image; model weights fetched inside the build, never via a laptop

Artifact Registry

One image with baked-in weights, shared by the service and the Job

IAM service account

Least-privilege runtime account for the bucket, the secret and the Job

NEXT ON THE ROADMAP

Document AI

Title deeds, valuation reports and repair invoices cross-checked against the images

Vertex AI Search

Policy wording and lending guidelines retrieved for the assistant

Vertex AI + BigQuery ML

Claim-likelihood and collateral-risk models trained on the inspection history

Cloud SQL / AlloyDB

Multi-tenant case management, audit trail and SSO

RESPONSIBLE AI

Compliant by design: humans decide, AI shows evidence

Human in the loop

Decision support only: a qualified surveyor or underwriter makes the final call.

Explainable

Deterministic YAML rules, and every decision lists its reasons and risk points.

Auditable

Raw model outputs are archived, each inspection is a JSON record in Cloud Storage, and the config is versioned.

Safe wording

No claims about structural safety or root cause. Locks, fans and windows always become manual checks.

Secure

Least-privilege service account, Gemini key in Secret Manager, no keys in files, and data stays in the client's own project.

Licence-aware

LocateAnything weights are non-commercial; Gemini localisation is the drop-in swap for production.

BUSINESS MODEL

Priced per inspection, far below a site visit

Pay-per-inspection API

Lenders and insurers send a walkthrough video from their own apps and get JSON, a report and a decision back.

Enterprise subscription

Portfolio dashboard, BigQuery export, custom risk rules, SSO, and private deployment in the client's Google Cloud project.

Inspector & valuer firms

White-label app: the AI drafts every report and coverage control, so each inspector handles more cases.

Unit economics (indicative)

Gemini calls

≈ $0.005 each → ≈ $0.35 per walkthrough

L4 GPU Job

≈ $0.50 of L4 time per 20-min walkthrough; no idle GPU

Video inspection

≈ $0.85 per walkthrough, L4 Job + Gemini

Idle footprint

≈ $1.20 per month (image storage); $30/month budget alert

Pricing hypothesis to validate in pilots: a per-inspection fee well below a physical visit, with volume tiers for portfolios.

GO-TO-MARKET & PITCH

Start with housing finance and home insurance in APAC

PILOT — 0–3 months

1 housing-finance lender + 1 home insurer. Run AI alongside existing inspections on ~500 cases and measure turnaround, cost and referral accuracy.

INTEGRATE — 3–6 months

Pub/Sub into the loan-origination and claims systems, BigQuery portfolio dashboards, and custom risk rules per client.

SCALE — 6–18 months

Google Cloud Marketplace listing, partnerships with valuer and inspector networks, and expansion to commercial property and claims.

Pilot KPIs: turnaround time · cost per inspection · % cases auto-cleared vs referred · fraud flags confirmed · re-inspection rate

> ***Every lender and insurer already asks for photos. We replace them with a short phone video that is trustworthy, explainable and ready for a decision.***

ROADMAP

Next: an inspection agent, then portfolio-level risk

NOW · v1.0.0

- Video walkthrough inspection
- Risk and valuation engine
- Fraud and coverage checks
- Gemini analysis and Q&A
- Cloud Run service + GPU Job

NEXT · 3 months

- Inspection agent built with Gemini on Agent Platform (ADK)
- Live capture guidance: tells the inspector what to film
- Spatial reasoning over rooms, surfaces, objects, findings
- Answers for every stakeholder, with human approval

LATER · 6–12 months

- BigQuery ML claim and collateral-risk scoring
- Before/after claim comparison in 3D
- Document AI and Vertex AI Search: deeds and policies
- Agent hand-offs to loan and claims systems

Goal: from one trusted inspection to a portfolio-level risk signal for every loan and every policy.

THANK YOU

Trustworthy property evidence for every loan and every policy

Minutes, not days

A phone video becomes a decision-ready risk file.

Fraud-resistant

Recycled videos and screen or AI footage are caught, and coverage is proven in 3D.

Explainable

LocateAnything and Gemma 4 find, Gemini explains, transparent rules decide, and humans sign off.







Prototype v1.0.0 · one-command Cloud Run deploy · Gemini · Gemma · Cloud Storage